Statistics 652: Fall 2026
Problem Set 4 *Note: Additional problems may be added to this problem set.*

4.1. Suppose $\mathbf{X} = (X_1, X_2)$ has the joint distribution function

$$F(x_1, x_2) = \begin{cases} 1 & x_1 \geq 1, x_2 \geq 1 \\ min(x_1, x_2) & 0 \leq x_1 \leq 1, x_2 \geq 0 \text{ or } 0 \leq x_2 \leq 1, x_1 \geq 0 \\ 0 & x_1 < 0 \text{ or } x_2 < 0 \end{cases} .$$

i) Show that F is continuous everywhere, but that $\frac{\partial^2 F(x_1,x_2)}{\partial x_1 \partial x_1} = 0$ almost everywhere, i.e. except on a set of area zero in R^2.

ii) Show that $F(x_1, x_2)$ corresponds to the joint distribution function for the uniform distribution on the line connection $(0, 0)$ to $(1, 1)$.

4.2. Show that if $\mathbf{A}$ and $\mathbf{B}$ are symmetric matrices of order q such that $y\prime\mathbf{A}y = y\prime\mathbf{B}y$ for all $y \in \mathrm{R}^q$, then $\mathbf{A} = \mathbf{B}$.

4.3. Referring to class Lecture 11:

i) Verify the properties of an orthogonal projection $\mathbf{Q}$ given at the bottom of page 4.

ii) Verify the note regarding g-inverses at the top of page 6.

iii) Prove Lemma 11.9, the multivariate WLLN and the multivariate CLT given at the top of page 7.

iv) Referring to Lemma 11.11, give an example for which $\mathbf{\Sigma}_k \to \mathbf{\Sigma}$ but $rank\{\mathbf{\Sigma}_k\} \not\to rank\{\mathbf{\Sigma}\}$.

4.4. Referring to class Lecture 12:

i) Show $rank\{\mathbf{\Sigma}_\pi\} = q - 1$, where $\mathbf{\Sigma}_\pi = diag\{\pi\} - \pi\pi'$ as defined on page 2.

ii) If $b \in \mathrm{R}^q$ with $b'b = 1$, show that $(I_q - bb')^- = (I_q - bb')$.

4.5. Referring to class Lecture 13: Suppose $h_n : \mathrm{R}^p \to \mathrm{R}$, for $n = 1, 2, \ldots$, and $h : \mathrm{R}^p \to \mathrm{R}$ are continuous and bounded above with $h(y)$ having a unique maximum at $y = y_o$. Show that if $h_n \to h$ point-wise as $n \to \infty$ then $y_n = argsup_y\{h_n(y)\} \to y_o$ as $n \to \infty$.

4.6. Regression. Let $Y \in \mathrm{R}$ and $\mathbf{X} \in \mathrm{R}^p$.

i) Show that $E[(Y - g(\mathbf{X}))^2]$ is minimized if and only if $g(X) = E[Y|\mathbf{X}]$.

ii) For $\epsilon = Y - E[Y|\mathbf{X}]$, show that $Cov(h(\mathbf{X}), \epsilon) = \mathbf{0}$ for any function $h : \mathrm{R}^p \to \mathrm{R}^q$.

iii) Show that $E[var(Y|\mathbf{X})] = var(\epsilon)$.

4.7. Linear Regression. Let $Y \in \mathrm{R}$ and $\mathbf{X} \in \mathrm{R}^p$, with $\mu_Y = E[Y], \boldsymbol{\mu}_\mathbf{X} = E[\mathbf{X}]$, $\sigma_{YY} = \sigma_Y^2 = var(Y), \mathbf{\Sigma}_{\mathbf{XX}} = Var(\mathbf{X})$ and $\boldsymbol{\sigma}_{\mathbf{X}Y} = Cov(\mathbf{X}, Y)$.

i) Show that $E[(Y - \alpha - \boldsymbol{\beta}^T\mathbf{X})^2]$ is minimized if and only if $\boldsymbol{\beta} = \mathbf{\Sigma}_{\mathbf{XX}}^{-1}\boldsymbol{\sigma}_{\mathbf{X}Y}$ and $\alpha = \mu_Y - \boldsymbol{\beta}^T\boldsymbol{\mu}_\mathbf{X}$.

ii) Consider the linear regression of Y on $\mathbf{X}$, i.e. $\hat{Y} = \alpha + \boldsymbol{\beta}^T\mathbf{X}$, with α and $\boldsymbol{\beta}$ defined as in part (i). The error due to linear regression is defined as $\epsilon = Y - \widehat{Y}$. Show that $Cov(\mathbf{X}, \epsilon) = 0$ and $var(Y) = var(\widehat{Y}) + var(\epsilon)$.

4.8. Multivariate Linear Regression. , with $\boldsymbol{\mu}_{\mathbf{Y}} = E[\mathbf{Y}]$, $\boldsymbol{\mu}_{\mathbf{X}} = E[\mathbf{X}]$, $\boldsymbol{\Sigma}_{\mathbf{YY}} = var(\mathbf{Y}), \boldsymbol{\Sigma}_{\mathbf{XX}} = Var(\mathbf{X})$ and $\boldsymbol{\Sigma}_{\mathbf{XY}} = cov(\mathbf{X}, \mathbf{Y})$. Define the multivariate linear regression of $\mathbf{Y}$ on $\mathbf{X}$ as $\widehat{\mathbf{Y}} = \boldsymbol{\alpha} + \mathbf{B}^T\mathbf{X}$ where $\mathbf{B} = \boldsymbol{\Sigma}_{\mathbf{XX}}^{-1}\boldsymbol{\Sigma}_{\mathbf{XY}}$ and $\boldsymbol{\alpha} = \boldsymbol{\mu}_{\mathbf{Y}} - \mathbf{B}^T\boldsymbol{\mu}_{\mathbf{X}}$.

i) Show that the linear regression of Y_j on $\mathbf{X}$ is given by $\widehat{Y}_i$, and more generally that the linear regression of $\mathbf{a}^T\mathbf{Y}$ on $\mathbf{X}$ is given by $\mathbf{a}^T\widehat{\mathbf{Y}}$.

ii) Using part (i) and problem **4.2**, show that $E[(\mathbf{Y} - \boldsymbol{\alpha} - \mathbf{B}^T\mathbf{X})(\mathbf{Y} - \boldsymbol{\alpha} - \mathbf{B}^T\mathbf{X})^T]$ is minimized with respect to the partial ordering of symmetric matrices if $\boldsymbol{\alpha}$ and $\mathbf{B}$ are define as above.

iii) For $\boldsymbol{\epsilon} = \mathbf{Y} - \widehat{\mathbf{Y}}$, show that $Var(\boldsymbol{\epsilon}) = \boldsymbol{\Sigma}_{\mathbf{YY}} - \boldsymbol{\Sigma}_{\mathbf{XY}}^T\boldsymbol{\Sigma}_{\mathbf{XX}}^{-1}\boldsymbol{\Sigma}_{\mathbf{XY}}$. Also, show $Cov(\mathbf{X}, \boldsymbol{\epsilon}) = \mathbf{0}$ and $Var(\mathbf{Y}) = Var(\widehat{\mathbf{Y}}) + Var(\boldsymbol{\epsilon})$.

4.9. Simpson's paradox and partial correlations. For $\mathbf{Y} \in \mathrm{R}^{\mathrm{q}}$ and $\mathbf{X} \in \mathrm{R}^{\mathrm{p}}$, the partial correlation matrix of $\mathbf{Y}$ given $\mathbf{X}$, denoted $\mathbf{R}_{\mathbf{Y}\cdot\mathbf{X}}$, is defined to be the correlation matrix obtain from $Var(\boldsymbol{\epsilon})$ as defined in **4.4.iii**.

i) Give a numerical example for which the correlation $\rho(Y_1, Y_2) > 0$ but the partial correlation $\rho(\{Y_1, Y_2\} \cdot X) < 0$.

ii) Give an example for which the conditional correlation $\rho(Y_1, Y_2|X) > 0$ but the partial correlation $\rho(\{Y_1, Y_2\} \cdot X) < 0$.

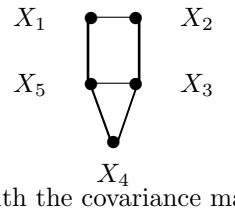
4.10. For $\mathbf{y} \in \mathrm{R}^{\mathrm{n}}$ and subspace $\mathcal{V} \subset \mathrm{R}^{\mathrm{n}}$, show that $\min\{\|\mathbf{y} - \widehat{\mathbf{y}}\| \mid \widehat{\mathbf{y}} \in \mathcal{V}\}$ is uniquely obtained when $\widehat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto $\mathcal{V}$.

4.11. Show that if $\mathbf{Q}$ is an orthogonal projection, then its eigenvalues are either 1 or 0, and state their multiplicities. Also, show $\mathrm{rank}(\mathbf{Q}) = \mathrm{trace}(\mathbf{Q})$.

4.12. For a random vector $\mathbf{X} \in \mathrm{R}^{\mathrm{q}}$ with covariance matrix $\boldsymbol{\Sigma}$, show that if $\sigma^{jk} \equiv \{\boldsymbol{\Sigma}^{-1}\}_{jk} = 0$ then the partial correlation between X_j and X_k given $\{X_i, i \notin \{j, k\}\}$ is zero.

4.13. Show that the factor or reduced-rank, and the factor analysis models for $\boldsymbol{\Sigma}$ are not convex in $\boldsymbol{\Sigma}^{-1}$.

4.14. i) Express the model for $\boldsymbol{\Sigma}^{-1}$ associated with the graph:



ii) Construct the graph associated with the covariance matrix: $\boldsymbol{\Sigma} = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 3 \end{bmatrix}$.

4.15. For $\boldsymbol{\Sigma} \geq 0$, show that the largest eigenvalue $\lambda_1(\boldsymbol{\Sigma})$ is a convex function of $\boldsymbol{\Sigma}$ and that the smallest eigenvalue $\lambda_q(\boldsymbol{\Sigma})$ is a concave function of $\boldsymbol{\Sigma}$. Also show that the set $\mathcal{M} = \{\boldsymbol{\Sigma} > \mathbf{0} \mid \lambda_q(\boldsymbol{\Sigma}) \geq c\}$ is a convex set in $\boldsymbol{\Sigma}^{-1}$.

4.16. Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be a random sample from a q-dimensional random vector with covariance matrix $\boldsymbol{\Sigma}$, and let $\mathbf{S}$ be the unbiased sample covariance matrix. Use Jensens inequality to show $E[\lambda_1(\mathbf{S})] > \lambda_1(\boldsymbol{\Sigma})$ and $E[\lambda_q(\mathbf{S})] < \lambda_q(\boldsymbol{\Sigma})$, where $\lambda_1(\cdot)$ and $\lambda_q(\cdot)$ denote the largest and smallest eigenvalue function respectively.

4.17. Minimize over $\boldsymbol{\Sigma} > \mathbf{0}$ the penalized negative log-likelihood function $L(\boldsymbol{\Sigma}; \mathbf{S}_n) + \eta\Pi(\boldsymbol{\Sigma})$, where $L(\boldsymbol{\Sigma}, \mathbf{S}_n) = \mathrm{trace}(\boldsymbol{\Sigma}^{-1}\mathbf{S}_n) + \log\det(\boldsymbol{\Sigma})$, $\Pi(\boldsymbol{\Sigma}) = \mathrm{trace}(\boldsymbol{\Sigma}^{-1}) + \log\det(\boldsymbol{\Sigma})$, and $\eta \geq 0$. Report both the minimizer and the minimum.